



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

**CANDIDATE
NAME**

CLASS

CHEMISTRY

Preliminary Examination
Paper 2 Structured Questions (SPA)

9647/02
20 August 2015
2 hours

Candidates answer on the Question Paper.
Additional Materials: Data Booklet.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.

Answer **all** questions in the space provided.
You are advised to spend **30 minutes** on question 1.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

Answer **all** questions

- 1** The Solvay process is a major industrial method for the production of soda ash. As the ingredients for this process are readily available and inexpensive, Solvay-based chemical plants produce almost three-quarters of the supply worldwide.

In the first step of the manufacturing process, sodium bicarbonate, NaHCO_3 , is initially yielded. The reaction involves carbon dioxide gas passing through concentrated aqueous brine solution and ammonia gas:



Solid sodium bicarbonate is then precipitated out and filtered from the hot ammonium chloride solution.

Quality control tests on sodium bicarbonate sample **X** found trace amounts of contaminants including by-products and fragments from the alloy metal pipelines.

- (a)** You are required to design an experiment to determine the percentage purity of sodium bicarbonate in sample **X**. In your plan, you will collect and measure the volume of carbon dioxide gas produced from the reaction between a known quantity of baking soda and sulfuric acid:



You are provided with:

- Sample **X**
- $0.200 \text{ mol dm}^{-3}$ sulfuric acid
- All common apparatus normally found in a school laboratory

In your plan you should:

- include clear calculations on the mass and volume of the reactants to be used in your experiment, stating any assumptions made
- draw a fully-labelled diagram of the assembled apparatus you would use
- give a full description of the essential experimental procedures

Diagram:

(b) State one precaution that should be taken to ensure the reliability of the experiment.

Please turn over]

- (c) The contaminants were suspected to contain the cations Pb^{2+} and Zn^{2+} as well as the by-product NH_4Cl .

Outline a series of test-tube experiments in order to identify the presence of these ions. Your plan should include all chemical tests, expected observations and deductions.

Identification by elimination method is not allowed in your plan.

Equations are not required in your plan.

Step	Test	Observation	Deduction

[5]

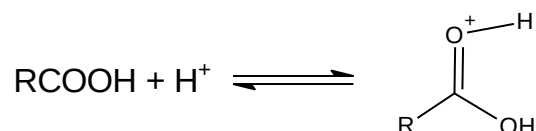
[Total : 12]

Please turn over]

- 2 Esters are derivatives of carboxylic acids. Simple esters tend to have pleasant, fruity odors and are widely used as flavours and fragrances. Esterification generally refers to the formation of esters from alcohol and carboxylic acids, as shown in **Equation 1**. This esterification mechanism is thought to involve 5 steps.



Step I : The carbonyl carbon on the carboxylic acid gains a proton.



Step II : Next, the electron-rich oxygen atom of the alcohol attacks the carbon atom of the carboxylic acid to form a positively-charged tetrahedral intermediate.

Step III : This is followed by transferring a proton to the hydroxyl oxygen of the carboxyl group.

Step IV : Subsequently, water is removed to form the positively charged ester.

Step V : The ester is formed when a proton is transferred to the base.

- (a) (i) Suggest why protonation in **Step I** is required before nucleophilic attack in **Step II** can occur.

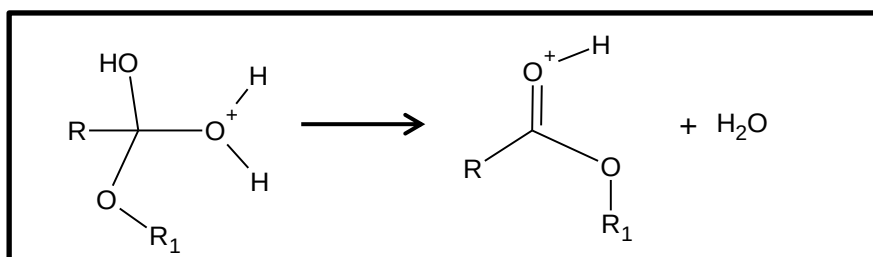
- (ii) Using the information given on page 5 to propose the mechanism for **only steps II and IV** in the given boxes below.

In your answer, you should include all charges and lone pairs and show the movement of electrons by curly arrows.

Step II :



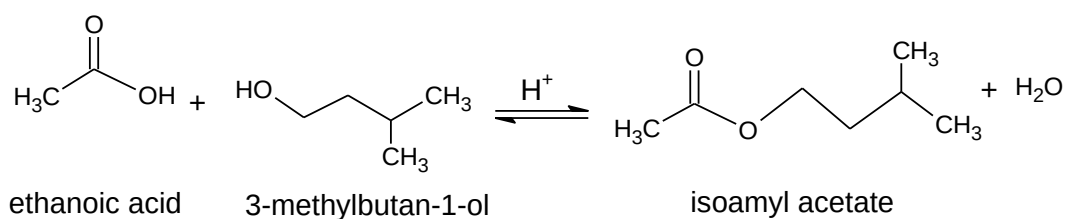
Step IV :



[3]

- (b) Following the success of the movie, *The Minions*, food manufacturers have sought to include banana flavours in their products. The compound that gives characteristic banana flavour is isoamyl acetate.

Isoamyl acetate can be synthesised via esterification as follows :



Data about the compounds is given below.

Compound	Boiling point/ °C	Density/ g cm ⁻³	M _r	Solubility in water
3-methylbutan-1-ol	130	0.809	88.0	slightly soluble
ethanoic acid	118	1.05	60.0	soluble
concentrated sulfuric acid	330	1.84	98.0	soluble
water	100	1.00	18.0	-
isoamyl acetate	142	0.876	130.0	slightly soluble

Synthesis of crude isoamyl acetate

1. Add 3.2 g of 3-methylbutan-1-ol into a round bottom flask. Add 5 cm³ of ethanoic acid and 3 boiling chips into the flask.
2. Add 6 drops of concentrated sulfuric acid.
3. Construct a reflux set up and heat the mixture for 30 min.
4. At the end of the reflux period, remove the flask from the heat and let the mixture cool to room temperature.

- (i) Using the information given, determine, by calculation, which reagent is in excess.

- (ii) Draw a reflux setup, indicating the flow of water clearly.

[3]

(c)

Purification of **crude** isoamyl acetate

5. Shake the crude isoamyl acetate with 25 cm³ water in a separating funnel and separate the aqueous layer from isoamyl acetate. Reject the aqueous layer.
6. Return the isoamyl acetate to the funnel, cautiously add dilute aqueous sodium carbonate in the separating funnel, releasing the pressure at intervals.

- (i) Give **two** reasons why the addition of water in **step 5** is necessary before step 6.

- (ii) In **step 6**, why is it important to release the pressure at intervals?

[3]

(d)

Purification of **crude** isoamyl acetate

7. Transfer the isoamyl acetate into a conical flask and add some granular anhydrous sodium sulfate. Swirl the mixture until the liquid is clear.
8. Filter the isoamyl acetate into a clean, dry round-bottomed flask and distil it. Collect the fraction by boiling over a suitable range.

- (i) What is the role of anhydrous sodium sulfate?

- (ii) Explain the significance of a clear liquid obtained after **step 7**.

- (iii) With reference to **steps 5 to 8**, suggest why the reactant you have determined in **(b)(i)** must be in excess.

[3]

- (e) After separating isoamyl acetate from water, there will still be a very small amount of one of the original reactants present as an impurity. To remove this, the isoamyl acetate is shaken with reagent **X**. A reaction occurs with the impurity present in the isoamyl acetate, increasing its solubility.

- (i) Suggest the identity of the **impurity** and reagent **X**.

Impurity: _____

Reagent **X**: _____

- (ii) By using structural formula, write a chemical equation for this reaction.

- (iii) Suggest why would adding reagent **X** increase the solubility of the impurity to aid its removal.

[3]

- (f) After the distillate was collected, it was discovered that the boiling point is 10 °C higher than the expected temperature.

- (i) Give a possible reason for the temperature difference.

- (ii) Outline another technique to obtain **pure** isoamyl acetate other than distillation.

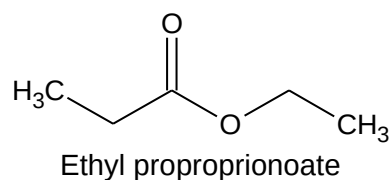
[3]

- (g) After purification, the mass of isoamyl acetate obtained is 2.77g. Calculate the percentage yield of this synthesis.

[1]

- (h) A bottle containing the product, isoamyl acetate, was wrongly labelled as ethyl propanoate.

Describe a simple chemical test to distinguish between the two bottles, clearly stating the observation(s) made. Include balanced chemical equations for your chosen test.



Test:

Observation:

Equations:

[4]

- (i) Using your chemical knowledge of the reactivity of acyl chlorides, suggest why ethanoyl chloride would be a better alternative to make isoamyl acetate.

[2]

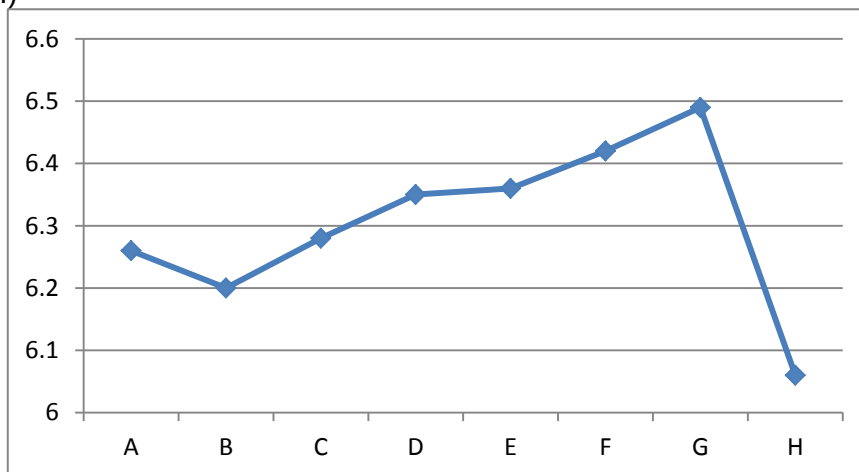
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- 3 The periodic table is a tabular arrangement of the chemical elements, organised by their atomic number, electronic configurations and recurring chemical properties. Dimitri Mendeleev is generally credited with the first publication of the first widely recognised Periodic Table in 1869.

Elements **A**, **B**, **C**, **D**, **E**, **F**, **G** and **H** are consecutive elements in the Periodic Table from periods 3 to 4.

The second ionisation energy (I.E.) of elements **A** to **H** are illustrated below.

Log (2^{nd} I.E.)



- (a) (i) With the aid of a relevant equation, explain what is meant by the term **second ionisation energy** of the element **D**.

- (ii) Account for the difference in second ionisation energy of

1. **A** and **B**

2. D and E.

[5]

- (b)** By interpreting the data above, deduce the identity of elements **B** and **F** and compare their melting points using structure and bonding.

[2]

- (c)** When dissolved in water, the chlorides of elements **B** and **C** produce solutions that turn blue litmus red. Explain.

[1]

- (d) Oxides of **A**, **B** and **H** forms white solids. Suggest how they may be differentiated from one another. Write balanced chemical equations for any reactions that occur.

[3]

- (e) (i) Describe what you would see when element **D** is burned in air. Write equations for the reactions that occur.

Observation: _____

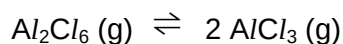
Equation: _____

- (ii) The oxides of **D** and **G** resulting from the reactions in (e)(i), both react with water. Write balanced equations for these two reactions and describe the colours observed when methyl orange is added to the resulting solutions.

[4]

[Total: 15]

- 4 Aluminium chloride is a white compound that is used in the production of aluminium metal and in chemical industries. It exists as a dimer in liquid state and starts to vapourise at a temperature of 120°C. At even higher temperatures, the dimers dissociate into monomers as shown in the equation below.



- (a) Draw the displayed formula of the vapour at 120 °C, to show its bonding.

[1]

- (b) 2.63g of aluminium chloride was placed in a 300 cm³ vessel and heated to 227 °C. The pressure inside the flask rose from 1.36 atm to 1.7 atm.

- (i) Calculate the average relative molecular mass of the mixture at 227 °C.

- (ii) Hence, calculate the partial pressures of AlCl_3 and Al_2Cl_6 in the mixture.

- (iii) Write an expression for K_p and calculate its value.

- (iv) Use your answer in (b)(i) and (b)(iii) to sketch a graph to show how the average M_r of the mixture would vary with pressure starting from 1.7 atm.



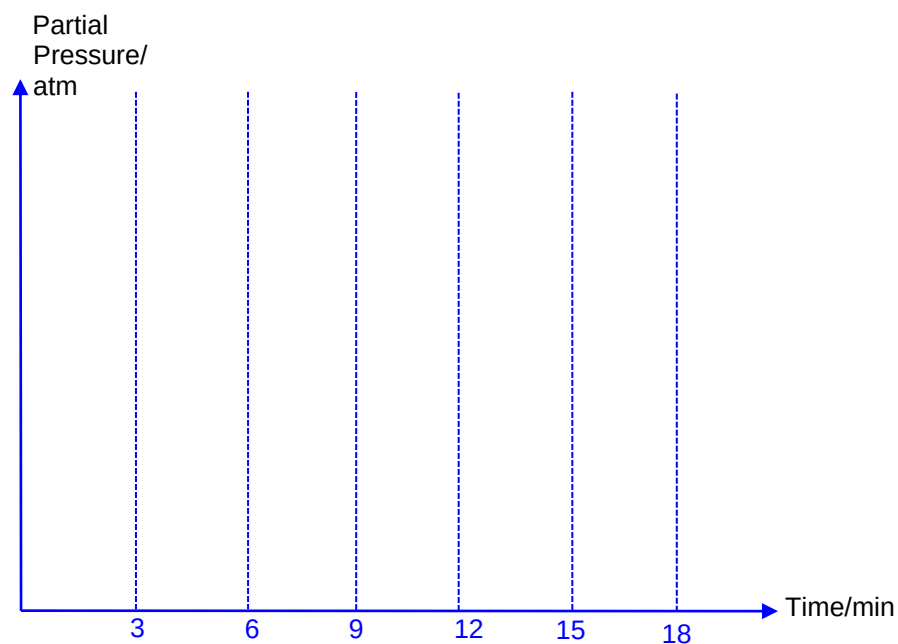
- (v) State and explain if the forward reaction is exothermic or endothermic.

- (vi) Hence, illustrate in a pressure-time graph, the changes in partial pressures of Al_2Cl_6 and $AlCl_3$ when

- the above equilibrium reaches equilibrium at the 3rd minute,
- temperature decreases to 150 °C at the 6th minute.
- partial pressure Al_2Cl_6 increases to 1.5 atm at the 12th minute.
- addition of inert gas into the vessel at the 18th minute.

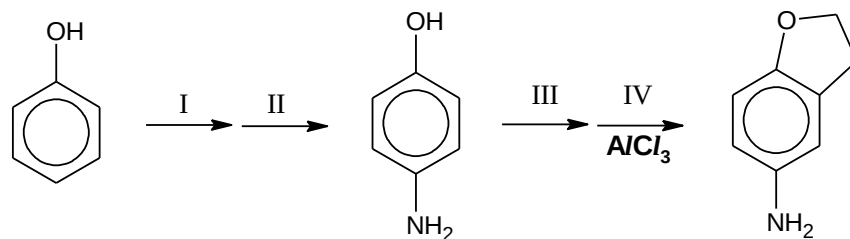
You may assume for **each** change, the system will take about 3 minutes to re-establish the equilibrium again.

[12]



Please turn over]

- (c) In the pharmaceutical industry, aluminium chloride is used as a catalyst in reactions such as Friedel-Crafts alkylation to attach an alkyl group onto a benzene ring. One synthesis of an organic compound involving AlCl_3 is shown below.



- (i) State the reagents and conditions for steps I and II.

Step I : _____

Step II : _____

- (ii) State the type of reaction in steps III and IV.

Step III : _____

Step IV : _____

- (iii) State the reacting species used in step III.

- (iv) Explain why aluminium chloride can act as a Lewis acid in step IV.

- (v) Suggest why AlCl_3 would make a better Lewis acid than AlBr_3 . [7]

[Total : 20 m]

END OF PAPER 2